
Guarded Block Transfers for Polytope Distance: Linear Convergence with Explicit Constants and Gap-Certified Screening for SVM Training

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Abstract

Hard-margin SVM training is the distance problem between two convex hulls, and the Triangle Algorithm (TA) solves it with a certified duality gap from supporting hyperplanes. We study the weight-transfer steps used by the enhanced TA and propose *guarded block transfers*: k Mitchell–Demyanov–Malozemov (MDM) transfers per scan are aggregated into a single exact line search, compared in $O(1)$ time against the single best transfer, and replaced by an away step whenever the leading transfer is capacity-clipped. Analysing the method on the Minkowski-difference polytope, where the objective is 1-strongly convex and one pyramidal width governs the geometry, we prove global linear convergence with explicit constants and without the combinatorial swap-step bounds of pairwise Frank–Wolfe. Because the TA’s certified gap is the Frank–Wolfe gap over that polytope divided by the current distance, gap-safe screening applies, and we give an explicit activation theorem: screening removes every zero-weight non-support point after $O\left(\frac{k}{\rho} \log \frac{1}{\tau}\right)$ iterations, where ρ is the linear rate constant and τ the optimal-face score margin. Every inequality in the analysis has a numerical check. On five LIBSVM benchmarks the solver matches the baselines’ solutions to five digits wherever both converge. It wins or ties on sparse-support cells (gisette’s linear, kernel and L_2 margins; kernel L_2 margins with expensive columns) against LIBSVM and kernel sequential minimal optimisation (SMO), and it is one to two orders of magnitude slower on dense-support L_2 margins and nearly touching kernel hulls, where LIBLINEAR finishes in seconds. We report both sides.

1 Introduction

Training a linear hard-margin SVM is equivalent to finding the closest points of the two classes’ convex hulls: the optimal separator is the perpendicular bisector of the witness pair, and the margin is half the hull distance [8]. Kalantari’s Triangle Algorithm (TA) exploits this *distance duality* directly [1]. It comes in two forms: TA I decides whether the hulls intersect, and TA II computes their distance, maintaining an iterate pair (p, q) and a certified interval $[LB, UB]$ on the distance from parallel supporting hyperplanes, so that it can stop at a provable relative accuracy ε . The enhanced TA (ETA) of [2] adds joint segment updates, dot-product caching, an anti-zig-zag heuristic and prioritised search, and reports large speedups over SMO on synthetic data.

The workhorse step of such geometric solvers is the *weight transfer*: move mass from the worst active vertex to the best candidate vertex, the Mitchell–Demyanov–Malozemov (MDM) step [4], which is exactly the pairwise Frank–Wolfe (FW) step of [3] on the relevant polytope; MDM converges linearly [22], and FW methods have a long history in SVM training [18, 20, 21]. Each step

costs an $O(n)$ scan and produces one transfer; we ask how much of the scan can be reused across several transfers without weakening the guarantee.

Contributions.

1. **Guarded block transfers** (§3): pair the top- k receivers with the worst- k active donors, take one exact line search along the aggregated direction, and *guard* by an $O(1)$ comparison with the single MDM step; on capacity-clipped iterations, fall back to an away step. Cache updates for a whole block are one matrix–vector product.
2. **Linear convergence with explicit constants** (Theorem 1): on the Minkowski-difference polytope $Z = \text{conv}(V) - \text{conv}(W)$ the guarded block algorithm inherits the pyramidal-width rate of pairwise FW up to a constant, and the away fallback replaces pairwise FW’s combinatorial swap-step bound with linear drop-step counting, as blended pairwise conditional gradients do with a different guard [12].
3. **Screening activation with explicit constants** (Theorem 2): the TA certificate is the FW gap divided by the distance (Lemma 3, an observation that goes back to Gilbert and Wolfe), so gap-safe screening [14] applies, and we bound the iteration by which it has removed every zero-weight non-support point in terms of the rate constant and the optimal-face margin.
4. **A regime study on real data** (§5). The analysis predicts $O(n^2)$ -type cost when the optimal support is dense (pyramidal width $\sim 1/n$). Two batteries on five LIBSVM datasets show the boundary from both sides: the solver wins or ties on sparse-support high-dimensional cells and loses by orders of magnitude on dense-support L_2 soft margins and on near-touching kernel hulls.

Code, proofs, numerical checks and experiment scripts are in the supplementary repository (anonymised for review).

2 Setting

Let $V = \{v_1, \dots, v_n\}$, $W = \{w_1, \dots, w_m\} \subset \mathbb{R}^d$. The polytope distance problem is

$$\min_{p \in \text{conv}(V), q \in \text{conv}(W)} \frac{1}{2} \|p - q\|^2. \quad (1)$$

Writing $z = p - q$ and $Z := \text{conv}(V) - \text{conv}(W) = \text{conv}\{v_i - w_j\}$, (1) is the minimum-norm-point problem $\min_{z \in Z} F(z)$, $F(z) = \frac{1}{2} \|z\|^2$, whose objective is 1-strongly convex and 1-smooth. Let z^* be the minimiser, $\delta^* = \|z^*\|$, $h(z) = F(z) - F(z^*)$, $M = \text{diam}(Z)$ and $\delta = \text{PW}(Z) > 0$ the pyramidal width [3].

The algorithm keeps simplex weights α on V , β on W , with $p = \sum \alpha_i v_i$, $q = \sum \beta_j w_j$; the induced product representation $\lambda_{ij} = \alpha_i \beta_j$ of z over the atoms $v_i - w_j$ has support $\text{supp } \alpha \times \text{supp } \beta$. With $g = -\nabla F(z) = q - p$, the *side scores* are $s_i^V = \langle g, v_i \rangle$ and $s_j^W = \langle p - q, w_j \rangle$; the implementation maintains them incrementally as differences of cached arrays Vp, Vq, Wp, Wq .

Certificate. For any iterate, $\text{UB} = \|z\|$ and $\text{LB} = \min_{u \in Z} \langle z, u \rangle / \|z\| = (\min_i \langle z, v_i \rangle - \max_j \langle z, w_j \rangle) / \text{UB}$ satisfy $\text{LB} \leq \delta^* \leq \text{UB}$; the TA stops when $\text{UB} - \text{LB} \leq \varepsilon \text{UB}$. The L_2 soft margin with regularisation parameter C , the ν -SVM and the kernel variants reduce to (1) exactly (augmented coordinates that turn the Gram matrix K into $K + I/C$; reduced convex hulls; feature-space images), see Appendix B.

Gaps. $g^{\text{FW}} = \max_{u \in Z} \langle g, u \rangle - \langle g, z \rangle$, $g^{\text{A}} = \langle g, z \rangle - \min_{u \in \text{supp } \lambda} \langle g, u \rangle$, $g^{\text{PW}} = g^{\text{FW}} + g^{\text{A}}$. Because $\langle g, v_i - w_j \rangle = s_i^V + s_j^W$ is separable and the support is a product set, each gap splits into a V -side and a W -side term computable from the score arrays in $O(n + m)$ (Lemma 4), and a transfer $v_r - v_u$ on the V side is a pairwise direction of Z (Lemma 5).

3 Guarded block transfers

Algorithm 1 amortises one $O(n)$ scan over up to k transfers. Only the scan is amortised: a new receiver still costs one Gram or kernel column, $O(nd)$, and a block touches up to $2k'$ columns, so

Algorithm 1 One iteration of the guarded block-transfer step (side V ; W symmetric)

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1: top- $k$  receivers by  $s^V$ ; worst- $k$  donors in  $\text{supp } \alpha$ ; pair best-with-worst:  $(r_j, u_j)$ ,  $j \leq k'$  ( $k' \leq k$ 
   pairs, limited by the support size); pair 1 is the MDM pair  $(r^*, u^*)$ 
2:  $d_j \leftarrow v_{r_j} - v_{u_j}$ ;  $\text{gap}_j \leftarrow s_{r_j}^V - s_{u_j}^V$ ;  $\gamma_j \leftarrow \min(\text{gap}_j / \|d_j\|^2, c_j)$   $\triangleright c_j$ : donor capacity
3: if pair 1 uncapped ( $\gamma_1 = \text{gap}_1 / \|d_1\|^2$ ) then
4:    $\Delta_B \leftarrow$  gain of exact line search  $t \in [0, 1]$  along  $d = \sum_j \gamma_j d_j$ ;  $\Delta_1 \leftarrow$  gain of the MDM
   step
5:   take the step with the larger gain  $\triangleright$  guard,  $O(1)$ 
6: else
7:   if the away step on  $u^*$  is boundary-clipped then take it  $\triangleright$  drop step:  $\alpha_{u^*} \rightarrow 0$ 
8:   else take the best of {block, MDM, away( $u^*$ ), toward( $r^*$ )}
9:   end if
10: end if

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81 Section 5 reports iterations, column evaluations and wall-clock separately. The implementation steps
 82 both sides per iteration, larger pairwise gap first, and skips the second side when the first step is a
 83 drop (Lemma 7). Every candidate gain is available in $O(1)$ from cached quantities: with $S = \langle g, d \rangle$
 84 and $D = \|d\|^2$, the block gain is $tS - t^2 D/2$ at $t = \min(S/D, 1)$, where $S = \sum_j \gamma_j \text{gap}_j$ and
 85 $D = \gamma^\top \Gamma \gamma$ for the $k' \times k'$ Gram matrix Γ of the pair directions, read off the cached columns; the
 86 MDM, away and toward gains are the standard one-dimensional expressions. Throughout, the *gain*
 87 of a step is the decrease $h(z_t) - h(z_{t+1})$ in the objective. The cache update for the whole block is
 88 a single matrix–vector product with the net per-index coefficients. Feasibility is immediate: donors
 89 lose at most $t\gamma_j \leq c_j$, receivers only gain, and the weight sum is conserved.

90 **Safe screening.** With $r = \sqrt{\text{UB}^2 - \max(\text{LB}, 0)^2}$ and $v_{\min} = \arg \min_i \langle z, v_i \rangle$, a zero-weight
 91 point with $\langle z, v_i \rangle - \langle z, v_{\min} \rangle > r \|v_i - v_{\min}\|$ is outside the optimal support and is removed (sym-
 92 metrically on W with the max). Because the optimum’s support survives, the reduced problem has
 93 the same optimum and all subsequent lower bounds remain valid.

94 4 Analysis

95 Full proofs are in Appendix A; we state the results and the ideas.

96 **Lemma 1** (Block gain). *Let $b_j = \gamma_j \|d_j\|$. Then $S \geq \sum_j b_j^2$, $D \leq k' S$, and the realised block gain*
 97 *satisfies $\Delta_B \geq S/(2k') \geq \Delta_{\text{MDM}}/(2k')$.*

98 The proof is the triangle inequality on $\|d\|$ and Cauchy–Schwarz on $(\sum b_j)^2$; capacity clipping only
 99 helps. With the guard, every case-(a) iteration’s gain is $\max(\Delta_B, \Delta_1) \geq \Delta_1$: the guarded step never
 100 makes less progress than pairwise FW.

101 **Lemma 2** (Good-step progress). *Work on the side with the larger pairwise gap. On any non-drop*
 102 *iteration, gain $\geq \min(g^{\text{PW}^2}/(32M^2), g^{\text{PW}}/8)$.*

103 Case (a) uses $g_{\text{side}}^{\text{PW}} \geq g^{\text{PW}}/2$ and the exact line search bound; case (b) uses $g_{\text{side}}^{\text{PW}} = g_{\text{side}}^{\text{FW}} + g_{\text{side}}^{\text{A}}$,
 104 so one of the toward step (always feasible with step ≤ 1) or the interior away step realises at least a
 105 quarter of g^{PW} .

106 **Theorem 1** (Global linear convergence). *Run the guarded block-transfer algorithm with away fall-*
 107 *back, one side per iteration, any $k \geq 1$, from any vertex pair (initial support $s_0 \leq 2$), and write*
 108 *$h_t = h(z_t)$. Then every non-drop iteration satisfies $h_{t+1} \leq (1 - \rho)h_t$ with*

$$\rho = \frac{1}{16} (\delta/M)^2,$$

109 *and among the first T iterations the number of drop steps is at most $k T_{\text{good}} + s_0$, where T_{good} is the*
 110 *number of non-drop iterations; hence $h_T \leq h_0 (1 - \rho)^{(T-s_0)/(k+1)}$. The two-sided schedule of the*
 111 *implementation (larger-gap side first, second side skipped after a drop) satisfies the same statement*
 112 *with the same ρ and exponent $(T - s_0)/(2k + 1)$ (Lemma 7).*

The rate follows from Lemma 2 and the geometric strong-convexity bound $g^{\text{PW}} \geq \delta\sqrt{2h}$ (Lemma 6, from the pyramidal-width definition of [3] and 1-strong convexity), together with $g^{\text{PW}} \geq g^{\text{FW}} \geq h$ for the linear branch. The drop counting is the away-step argument: each drop removes one support index, non-drop steps add at most k , so drops are at most k times the good steps plus s_0 . There is no swap-step bound because case (b) never takes a capacity-clipped pairwise step: it takes an away step, which either contracts or drops.

Proposition 1 (Near-orthogonal speed-up). *If $|\langle d_i, d_j \rangle| \leq \eta \|d_i\| \|d_j\|$ for $i \neq j$, the interior block gain is at least $(\sum_j \Delta_j)/(1 + \eta(k' - 1))$, where Δ_j is pair j 's individual gain.*

As $\eta \rightarrow 0$ a single scan realises the sum of all k pairwise gains; in high dimension distinct support directions are nearly orthogonal, which is the mechanism behind the measured 4–30 $\times$ iteration reductions.

Lemma 3 (The certified gap is the FW gap). *At any full scan, $\text{UB} - \text{LB} = g^{\text{FW}}(z)/\text{UB}$, and $g^{\text{FW}}(z) \leq 2h + R\sqrt{2h}$ with $R = \max_{u \in Z} \|u\|$.*

Theorem 2 (Screening activation). *Assume $\delta^* > 0$, let F_V, F_W be the optimal faces (the vertices that can carry weight at the optimum), let $\tau_i = \langle z^*, v_i \rangle - \min_j \langle z^*, v_j \rangle$ be the score margin of v_i (and symmetrically on W), $\tau > 0$ the minimal nonzero margin, and L the larger class diameter. A zero-weight point with margin τ_i is screened as soon as $r \leq \tau_i/(3L)$; consequently, once $h_t \leq \tau^4 \delta^{*2}/(2592 R^4 L^4)$, which under Theorem 1 holds for all $t \geq s_0 + \frac{k+1}{\rho} \log(2592 R^4 L^4 h_0/(\tau^4 \delta^{*2}))$, every zero-weight point outside $F_V \cup F_W$ is removed permanently, and each later scan costs $O(|F_V| + |F_W| + |\text{supp } \lambda|)$ instead of $O(n + m)$.*

The τ -dependence is unavoidable: no rule that is safe for all instances consistent with a certificate of gap $\sim \tau$ can distinguish a point of margin τ from a support point. Each of Lemma 1, Proposition 1, Lemma 3, Lemma 7 and Theorem 2 is verified numerically on random and adversarial ensembles at machine precision (Appendix D).

5 Experiments

Setup. All solvers are validated against an exact quadratic program (QP; SLSQP over the two simplices) on small instances: distances agree to 10^{-9} or better for every variant. Stopping rule $\varepsilon = 10^{-3}$ on the certified relative gap throughout unless stated; baselines use matched tolerances on the same objective (Appendix C). Synthetic instances follow the protocol of [2] (two Gaussian clouds, $n = 5,000$ per class). Real data are the LIBSVM sets a9a, w8a, ijcnn1, gisette and covtype (100k subsample), standardised, with official test sets where they exist, five seeds, and a 600 s budget per solver and cell. All real-data runs share one dedicated 32-vCPU Xeon 2.6 GHz machine (122 GB, OpenBLAS, one BLAS thread per solver, 26 cells concurrently), so times are comparable within a battery. Every ETA-versus-baseline pair solves the *same* objective; a differently regularised solver appears only as a labelled reference. “Our SMO” is our own SMO implementation on the same objective; K-ETA is the kernelised ETA. The batteries were run with the unconditional V -then- W order that preceded the schedule of Lemma 7; the two differ only on drop iterations and agree on the instances of Appendix C.

Synthetic: steps, blocks, screening. Away and pairwise steps converge at $\varepsilon = 10^{-6}$ where the toward and midpoint steps of [2] do not, and block transfers cut iterations a further 4–30 $\times$ relative to single MDM; the wall-clock gain is smaller, 1.2–2.8 $\times$, because each block iteration touches more columns (Table 4). Screening reduces 10,000 points to 36–1,054 survivors and wall-clock by 1.4–2.4 $\times$ at $\varepsilon = 10^{-5}$, and is neutral at 10^{-3} , where runs end before the certificate activates, as Theorem 2 predicts. On seven synthetic and small-real problem classes the optimised solver is fastest to a certified solution on six (1.6–2.5 $\times$ ahead of LIBSVM, LIBLINEAR or NuSVC); the exception, MNIST odd-vs-even, is a dense-support L_2 problem. On the four classes where the single-MDM configuration also converges, blocks and guard add only 0.9–1.5 $\times$ in wall-clock; their clear wins are the two classes (tight tolerance, MNIST) where single MDM does not converge at all. Details and figures are in Appendix C.

Real data, sparse-support regime (Table 1). TA I first certifies linear separability: gisette is separable and a9a, w8a, ijcnn1 and covtype are not (hulls intersect, certified in 0.2–42 s), a diagnostic

Table 1: Regime battery on real data: hard margins and kernels, five seeds, 600 s per solver, mean $\pm$ 95% CI. Kernel cells (radial basis function kernel, RBF, $\gamma = 1/d$) use a class-balanced subsample of n_k training points. ETA and its baseline solve the same objective: LIN, linear hard margin (ETA with screening vs LIBSVM at $C=10^6$; our SMO in brackets); KHM, feature-space hard margin (K-ETA vs LIBSVM at $C=10^6$); KL2, kernel L_2 margin $K + I/C$, $C=1$ (K-ETA vs kernel SMO). conv = seeds in which ETA reached the certified gap (or certified intersection); oracle = $O(nd)$ column (ETA) or row (SMO) evaluations, in thousands.

cell	dataset	n_k / d	ETA (s)	baseline (s)	conv	dist. ETA / base	acc. ETA / base	oracle
LIN	gisette	6000 / 5000	56.6 $\pm$ 5.9	51.4 $\pm$ 13.1 [405 $\pm$ 62]	5/5	3.6638 / 3.6640	.981 / .981	1.5 [11.1]
KHM	gisette	6000 / 5000	86.2 $\pm$ 14.8	104.6 $\pm$ 20.2	5/5	.05215 / .05215	.979 / .979	2.6
KHM	ijcnn1	9853 / 22	421 $\pm$ 149	1.3 $\pm$ 0.3	4/5	.00430 / .00430	.946 / .946	2.5
KHM	a9a	10000 / 123	> 600	26.4 $\pm$ 2.9	0/5	$\leq$.00009 / .0033	.690 / .752	5.9
KHM	covtype	10000 / 54	> 600	144 $\pm$ 7 (maxiter)	0/5	$\leq$.00053 / .00025	.704 / .712	7.2
KHM	w8a	6479 / 300	101 $\pm$ 159	—	5/5	hulls intersect	—	1.9
KL2	gisette	6000 / 5000	131 $\pm$ 34	420 $\pm$ 49	5/5	.06882 / .06882	.971 / .971	3.6 / 15.2
KL2	w8a	6479 / 300	11.5 $\pm$ 2.6	41.7 $\pm$ 10.0	5/5	.05682 / .05682	.975 / .975	3.6 / 20.2
KL2	a9a	10000 / 123	52.8 $\pm$ 71.0	57.7 $\pm$ 23.2	5/5	.03019 / .03019	.795 / .795	7.6 / 43.6
KL2	covtype	10000 / 54	26.0 $\pm$ 4.2	17.6 $\pm$ 0.5	5/5	.02609 / .02609	.792 / .792	9.1 / 56.2
KL2	ijcnn1	9853 / 22	9.2 $\pm$ 2.5	6.1 $\pm$ 0.7	5/5	.04790 / .04790	.959 / .959	4.6 / 26.1

none of the baselines produces. On gisette ($d=5,000$; 1,233 of 6,000 points in the support) ETA ties LIBSVM on the linear hard margin, both $7\times$ ahead of our SMO; K-ETA is $1.2\times$ faster than LIBSVM on the RBF hard margin and $3.2\times$ faster than kernel SMO on the kernel L_2 margin. Distances agree to five digits, support counts within three, test accuracies exactly. Across all five KL2 cells K-ETA returns the same distance and accuracy as kernel SMO with 4–6 $\times$ fewer kernel-column evaluations; in wall-clock it is 3–4 $\times$ faster where columns are expensive (gisette, w8a) and 1.5 $\times$ slower where they are cheap (ijcnn1 $d=22$, covtype $d=54$), because the $O(n)$ NumPy scan per iteration then dominates. The reference hinge-loss SVC ($C=1$, a different objective) takes 1–84 s with accuracies within one point (Appendix C).

The KHM column shows where the geometry turns against the method. On ijcnn1 both solvers return the identical solution (distance 0.00430, 819 support vectors, accuracy 0.946), but K-ETA needs $1.5\text{--}2.5 \times 10^5$ iterations and 420 s against LIBSVM’s 1.3 s: the feature-space hulls are 0.004 apart in a space of diameter $\approx \sqrt{2}$, so δ/M in Theorem 1 is of order 10^{-3} , and LIBSVM’s second-order working-set selection with heuristic shrinking is far better suited to that conditioning. On a9a and covtype the hulls are within 10^{-4} and 5×10^{-4} of touching: K-ETA times out at gap 3–10, while LIBSVM’s $C=10^6$ surrogate stops at its Karush–Kuhn–Tucker (KKT) tolerance at a point 6–30 $\times$ farther out than K-ETA’s upper bound (or hits its iteration cap on covtype), so neither solver certifies anything, and the hard-margin kernel model is the wrong one there anyway (KL2 accuracies 0.79 vs 0.70–0.75). On w8a the feature-space hulls intersect (duplicate points with opposite labels); K-ETA certifies this in 101 s and no hard margin exists.

Real data, L_2 soft margins (Table 2). The same boundary appears within one problem class. Gisette is separable, so its L_2 margin has a sparse support (1,233 of 6,000 points) and ETA converges to the certified gap in 30–45 s at every C , $7\times$ faster than our SMO on the same objective. LIBLINEAR run to the same primal accuracy (tolerance 10^{-6} ; at $C=1$: 0.14900 vs ETA’s 0.14893, inside ETA’s certified 8×10^{-4}) takes 8 min to 1.5 h, but that is the tolerance needed to match ETA’s primal, not a property of the problem; we did not time LIBLINEAR at its default tolerance (Appendix C) and therefore claim no wall-clock advantage over it on gisette. What ETA adds is the certificate: at $C=10$ LIBLINEAR’s progress-based stopping rule fires, even at tolerance 10^{-6} , at a primal 5% above ETA’s certified optimum (0.1566 vs 0.1490, test accuracy 0.977 vs 0.981), because on separable data the primal is nearly flat at large C ; the certificate does not fire early. On the four overlapping datasets both geometric solvers hit the 600 s cap on every cell while LIBLINEAR finishes in 0.3–25 s. At $C=0.1$ ETA’s iterate already has LIBLINEAR’s test accuracy on a9a, ijcnn1 and w8a (0.849, 0.918, 0.987) when the cap is reached with the certificate still at 0.01–0.6: the iterate converges long before the certificate closes. At $C \geq 1$ the augmented hulls ($K + I/C$) approach each other, the certified gap at the cap exceeds one and the iterate is not yet usable; on covtype (100,000 points) it is not usable at any C . Uncapped, ETA does converge on the overlapping sets: on a9a ($C=0.1$, ten runs) in $2,908 \pm 138$ s and 95,201 iterations, with primal objective

Table 2: L_2 soft-margin battery: five seeds (three for covtype at $C=0.1$), 600 s per solver for ETA and our SMO, LIBLINEAR unbounded; mean $\pm$ 95% CI. “cap” = timed out in every seed. LIBLINEAR is run at tolerance 10^{-6} so that its primal matches ETA’s certified value where ETA converges; its default-tolerance behaviour is discussed in Appendix C. ETA’s certified relative gap is reported at its stopping point (mean [min, max]); accuracies are those of the returned iterate. Two covtype $C=0.1$ seeds were still running when this draft was prepared.

dataset (n / d)	C	ETA (s)	SMO (s)	LIBLINEAR (s)	ETA gap at stop	acc. ETA / SMO / LIBLINEAR
gisette (6,000 / 5,000)	0.1	35.3 ± 6.8	243 ± 53	500 ± 41	8×10^{-4}	.981 / .981 / .981
	1	43.3 ± 5.3	295 ± 6	3916 ± 115	8×10^{-4}	.981 / .981 / .981
	10	33.2 ± 4.6	236 ± 36	5396 ± 197	10×10^{-4}	.981 / .981 / .977
a9a (32,561 / 123)	0.1	cap	cap	3.7 ± 0.8	0.15 [0.10, 0.23]	.849 / .845 / .849
	1	cap	cap	6.8 ± 1.7	1.26 [1.25, 1.28]	.810 / .780 / .849
	10	cap	cap	2.3 ± 0.5	1.83 [1.80, 1.86]	.635 / .763 / .849
ijcnn1 (49,990 / 22)	0.1	cap	478 ± 181	0.3 ± 0.1	0.010 [0.005, 0.016]	.918 / .918 / .918
	1	cap	cap	0.6 ± 0.2	0.68 [0.61, 0.73]	.901 / .917 / .918
	10	cap	cap	0.4 ± 0.1	1.34 [1.27, 1.40]	.656 / .852 / .918
w8a (49,749 / 300)	0.1	cap	cap	22.7 ± 4.3	0.57 [0.50, 0.66]	.987 / .985 / .987
	1	cap	cap	20.6 ± 6.6	1.17 [1.14, 1.21]	.923 / .979 / .987
	10	cap	cap	13.2 ± 1.1	2.02 [1.94, 2.14]	.720 / .971 / .987
covtype (100,000 / 54)	0.1	cap	cap	14.7 ± 22.8	1.04 [1.03, 1.05]	.542 / .651 / .755
	1	cap	cap	6.1 ± 3.6	1.42 [1.23, 1.73]	.508 / .541 / .756
	10	cap	cap	5.0 ± 1.8	2.01 [1.20, 2.37]	.517 / .514 / .756

687.07, identical to LIBLINEAR’s to five digits, and on w8a in 5,153 s. Figure 1 (appendix) shows these traces: the linear-rate signature of Theorem 1, but with 60–65% of the points in the support and near-uniform weights, so that $PW(Z) \sim 1/n$ and the cost is $O(n^2)$ -type. Centroid initialisation and larger blocks did not help (the former increased the iteration count, the latter traded iterations for per-iteration cost).

6 Related work

Distance duality and the TA [1, 2]; MDM [4] and its linear rate [22]; Gilbert’s algorithm [5] and Wolfe’s method [23]; FW for polytope distance and SVMs through coresets [18, 19], core vector machines [20] and the FW–SVM line of [21]; the nearest-point and reduced-hull view of soft margins [6, 8]; away steps [7] and the pyramidal-width analysis [3], whose pairwise step is our MDM pair. *Swap steps*. Blended pairwise conditional gradients [12] also remove pairwise FW’s swap-step bound with a guard, there between the pairwise step and a FW step, and [13] avoid bad steps with a short-step chain. Our guard compares an aggregated block of k transfers with the single MDM step and falls back to an away step, so the drop-step count of away-step FW applies unchanged. Block-coordinate FW [9] blocks over *coordinates* of a product domain rather than aggregating pairs on one polytope. *Screening and identification*. Gap-safe rules [14, 15] screen with a radius from the duality gap and identify the support in finite time; the active-set complexity of away-step FW [16] and the strict-complementarity bounds of [17] give the same identification from the algorithm’s side. Theorem 2 is the polytope-distance instance, with constants tied to the TA certificate; the SVM rules of [10, 11] are its ancestors.

7 Conclusion

Guarded block transfers give the Triangle Algorithm a linear rate with explicit constants and no combinatorial swap-step bounds, together with a screening rule that is both safe and provably effective. The cost is a factor $(k + 1)$ in the exponent, recovered in the near-orthogonal regime where a block realises close to the sum of its k pairwise gains. The experiments locate the boundary the analysis predicts: sparse support and well-separated hulls, where the solver matches or beats LIBSVM and kernel SMO and supplies a certificate, against dense support or nearly touching hulls, where δ/M is small and primal coordinate descent or second-order SMO is faster by one to two orders of magnitude. Which side a run is on can be read from its certified gap and support size. Limitations: the implementation is NumPy-level (a compiled inner loop would move the practical boundary, not the $O(n^2)$ asymptotics), sparse inputs are not supported, the reduced-hull (ν -SVM) variant is validated but not analysed, and the constants in Theorem 1 are loose.

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AI assistance. AI tools were used in the preparation of the code, experiments and manuscript; the authors verified all results and are responsible for the content.

A Proofs

Lemma 4 (Separability). *With the product representation, $g^{\text{PW}} = g_V^{\text{PW}} + g_W^{\text{PW}}$ where $g_V^{\text{PW}} = \max_i s_i^V - \min_{i \in \text{supp } \alpha} s_i^V \geq 0$ and g_W^{PW} likewise; the same holds for g^{FW} and g^{A} .*

Proof. $\langle g, v_i - w_j \rangle = s_i^V + s_j^W$ is separable and the support is a product set, so the max over atoms and the min over the support separate; each side term is nonnegative since the support max dominates the support min and the global max dominates the support max. $\square$

Lemma 5 (Side transfers are Z -pairwise directions). *For $\alpha_u > 0$ and any $b \in \text{supp } \beta$, $v_r - v_u = (v_r - w_b) - (v_u - w_b)$ is a difference of an atom of Z and an atom in the support, and the transfer $\gamma \leq \alpha_u$ realises $z \leftarrow z + \gamma(v_r - v_u)$ inside Z .*

Proof of Lemma 1. Since $\gamma_j \leq \text{gap}_j / \|d_j\|^2$ we have $\gamma_j \text{gap}_j \geq \gamma_j^2 \|d_j\|^2 = b_j^2$, so $S = \sum_j \gamma_j \text{gap}_j \geq \sum_j b_j^2$. By the triangle inequality $\|d\| \leq \sum_j \gamma_j \|d_j\| = \sum_j b_j$ and by Cauchy-Schwarz $(\sum_j b_j)^2 \leq k' \sum_j b_j^2 \leq k' S$, hence $D \leq k' S$. Exact line search on the 1-smooth F gives $\Delta_B = S^2 / (2D) \geq S / (2k')$ in the interior case and $\Delta_B = S - D/2 \geq S/2$ at the boundary $t = 1$ (where $D \leq S$). Finally $S \geq \gamma_1 \text{gap}_1 \geq \Delta_{\text{MDM}}$ since all terms are nonnegative and $\Delta_{\text{MDM}} = \gamma_1 \text{gap}_1 - \gamma_1^2 \|d_1\|^2 / 2$. $\square$

Proof of Lemma 2. Let $g_S = \max(g_V^{\text{PW}}, g_W^{\text{PW}}) \geq g^{\text{PW}} / 2$. Every direction below is a difference of two points of one class hull, so its norm is at most $\text{diam}(\text{conv } V) \leq M$. For a feasible direction with directional derivative $\sigma > 0$, squared norm D and maximal step $\bar{t}$, exact line search on a 1-smooth function gives gain $\geq \min(\sigma^2 / (2D), \bar{t}\sigma/2)$ (interior optimum, or the clipped step evaluated with $\bar{t}D \leq \sigma$). Case (a): the guard's gain is at least the MDM gain, whose derivative is g_S and whose interior optimum is attained, so gain $\geq g_S^2 / (2M^2) \geq g^{\text{PW}2} / (8M^2)$. Case (b), non-drop: $g_S = g_{\text{side}}^{\text{FW}} + g_{\text{side}}^{\text{A}}$, so one term is $\geq g_S / 2 \geq g^{\text{PW}} / 4$. If it is the FW term, the toward step (maximal step 1) gives gain $\geq \min(g_{\text{side}}^{\text{FW}2} / (2M^2), g_{\text{side}}^{\text{FW}} / 2) \geq \min(g^{\text{PW}2} / (32M^2), g^{\text{PW}} / 8)$; if it is the away term, the away line search is interior (non-drop) and gain $\geq g_{\text{side}}^{\text{A}2} / (2M^2) \geq g^{\text{PW}2} / (32M^2)$. The algorithm takes the best available candidate. $\square$

Lemma 6 (Geometric strong convexity). *For every $z \in Z$ with a representation whose support contains an away atom, $g^{\text{PW}}(z) \geq \delta \sqrt{2h(z)}$ and $g^{\text{PW}}(z) \geq g^{\text{FW}}(z) \geq h(z)$.*

Proof. By the definition of the pyramidal width [3, Def. 3, Thm. 6], for $e = (z^* - z) / \|z^* - z\|$ the FW atom s and the away atom v satisfy $\langle g, s - v \rangle \geq \delta \langle g, e \rangle$. By 1-strong convexity, $\langle g, z^* - z \rangle \geq h + \frac{1}{2} \|z^* - z\|^2$, so with $\varrho = \|z^* - z\|$, $g^{\text{PW}} \geq \delta(h/\varrho + \varrho/2) \geq \delta \sqrt{2h}$ by AM-GM. Finally $g^{\text{PW}} = g^{\text{FW}} + g^{\text{A}} \geq g^{\text{FW}} = \max_u \langle g, u - z \rangle \geq \langle g, z^* - z \rangle \geq h$ by convexity. $\square$

Proof of Theorem 1. By Lemmas 2 and 6 a non-drop iteration gains at least $\min(2\delta^2 h / (32M^2), h/8) = (\delta/M)^2 h / 16$, using $\delta \leq M$; hence $h_{t+1} \leq (1 - \rho)h_t$. Drop counting: non-drop iterations add at most k support indices, drop iterations add none and remove at least one, and the support size stays ≥ 2 from s_0 ; so the number of removals, hence of drops, is at most $kT_{\text{good}} + s_0$, giving $T_{\text{good}} \geq (T - s_0) / (k + 1)$. All steps, drops included, are descent steps, so h is monotone. $\square$

Lemma 7 (Two-sided schedule). *Let every iteration compute both side gaps, take the guarded step of Algorithm 1 on the side with the larger pairwise gap first, and take the other side's guarded step only if the first step was not a drop; an iteration counts as a drop iff its first step is a drop. Then every non-drop iteration gains at least $\min(g^{\text{PW}2} / (32M^2), g^{\text{PW}} / 8)$ and $\# \text{drops} \leq 2kT_{\text{good}} + s_0$; hence Theorem 1 holds for this schedule with the same ρ and exponent $(T - s_0) / (2k + 1)$.*

Proof. On a non-drop iteration the first step is exactly the step analysed in Lemma 2, so it gains at least the stated amount; the second step is an exact line search along a feasible descent direction (or is skipped), so it cannot increase h . Drop counting: a drop iteration takes a single away step that removes one support index and adds none, because the second side is skipped; a non-drop iteration takes two guarded steps and adds at most k indices each. Removals, hence drops, therefore number at most $2kT_{\text{good}} + s_0$. $\square$

329 *Remark 1.* The skip is what makes the count work. If the second side always steps, an iteration
 330 whose analysed side drops can add up to k indices on the other side, and the number of such iterations
 331 is then bounded only by $\frac{k}{k+1}T$, which gives no rate for $k \geq 1$.

332 *Proof of Proposition 1.* $D = \sum_{i,j} \gamma_i \gamma_j \langle d_i, d_j \rangle \leq \sum_j b_j^2 + \eta \sum_{i \neq j} b_i b_j \leq (1 + \eta(k' - 1)) \sum_j b_j^2 \leq$
 333 $(1 + \eta(k' - 1))S$ by Cauchy–Schwarz on the cross terms; then $\Delta_B = S^2/(2D) \geq S/(2(1 + \eta(k' -$
 334 $1)))$, and for uncapped pairs $S = \sum_j 2\Delta_j$. $\square$

335 *Proof of Lemma 3.* $\text{LB} = \min_{u \in Z} \langle z, u \rangle / \|z\|$ by definition of the supporting hyperplane bound and
 336 $\text{UB} = \|z\|$; subtracting, $\text{UB} - \text{LB} = (\langle z, z \rangle - \min_u \langle z, u \rangle) / \|z\| = g^{\text{FW}} / \text{UB}$. At the optimum
 337 $0 = \langle z^*, z^* \rangle - \min_u \langle z^*, u \rangle$; subtracting from $g^{\text{FW}}(z)$, $g^{\text{FW}}(z) = (\|z\|^2 - \|z^*\|^2) + (\min_u \langle z^*, u \rangle -$
 338 $\min_u \langle z, u \rangle) \leq 2h + \max_u \langle z^* - z, u \rangle \leq 2h + R\|z - z^*\| \leq 2h + R\sqrt{2h}$ by 1-strong convexity. $\square$

339 *Proof of Theorem 2.* Strong convexity gives $\|z - z^*\| \leq r$ for the screening radius $r =$
 340 $\sqrt{\text{UB}^2 - \max(\text{LB}, 0)^2}$. Let $m^* = \min_j \langle z^*, v_j \rangle$ and pick $v' \in F_V$; since $v_{\min}$ minimises $\langle z, \cdot \rangle$,
 341 $\langle z^*, v_{\min} \rangle - m^* = \langle z, v_{\min} - v' \rangle + \langle z^* - z, v_{\min} - v' \rangle \leq rL$. Hence $\langle z, v_i - v_{\min} \rangle =$
 342 $\langle z^*, v_i - v_{\min} \rangle + \langle z - z^*, v_i - v_{\min} \rangle \geq \tau_i - rL - r\|v_i - v_{\min}\|$, and the rule fires when this exceeds
 343 $r\|v_i - v_{\min}\|$, for which $\tau_i \geq 3rL$ suffices. For the activation time, $r^2 = (\text{UB} - \text{LB})(\text{UB} + \text{LB}) \leq$
 344 $2R(\text{UB} - \text{LB}) = 2Rg^{\text{FW}}/\text{UB} \leq (2R/\delta^*)(2h + R\sqrt{2h}) \leq (4R^2/\delta^*)\sqrt{2h}$ for $h \leq R^2/2$
 345 (Lemma 3); requiring $r^2 \leq \tau^2/(9L^2)$ yields the stated condition on h_t , and Theorem 1 with
 346 $\log(1/(1 - \rho)) \geq \rho$ converts it into the iteration bound. Permanence follows from the safety of
 347 the rule. $\square$

348 B Exact reductions

349 **L_2 soft margin.** The problem $\min \frac{1}{2}\|w\|^2 + \frac{C}{2} \sum_i \xi_i^2$ is the hard-margin problem on the augmented
 350 points $\tilde{x}_i = [x_i; e_i/\sqrt{C}]$, whose Gram matrix is $K + I/C$ (K the Gram matrix of the original points)
 351 and whose augmented hulls never intersect; let δ_C denote their distance. The augmentation is never
 352 materialised: the extra coordinates of an iterate are exactly its convex weights, so every augmented
 353 inner product is a cached original-space product plus a sparse diagonal correction, $\langle \tilde{p}, \tilde{v}_i \rangle = \langle p, v_i \rangle +$
 354 α_i/C and $\|\tilde{p}\|^2 = \|p\|^2 + \|\alpha\|^2/C$. The separator is recovered as $w = 2(p - q)/\delta_C^2$, $b = (\|\tilde{q}\|^2 -$
 355 $\|\tilde{p}\|^2)/\delta_C^2$, and the primal value is $P = 2/\delta_C^2$. Validation: the solver agrees with an exact QP on the
 356 explicitly augmented problem to 10^{-10} across random instances and values of C ; the regularisation
 357 path δ_C is monotone in C ; on a9a ($C=0.1$) its certified primal 687.07 equals LIBLINEAR’s to five
 358 digits.

359 **ν -SVM.** The L_1 (hinge) soft margin is the nearest-point problem between *reduced* convex hulls
 360 $R(V, \mu) = \{\sum a_i v_i : \sum a_i = 1, 0 \leq a_i \leq \mu\}$, $\mu = 2/(\nu\ell)$ with $\ell = n + m$ the number of
 361 training points [8]. Two changes: the extreme-point oracle becomes a capped top- k blend (weight μ
 362 on the $\lfloor 1/\mu \rfloor$ best points in the search direction, via a partial sort, still $O(n)$), and the primary step is
 363 the capped MDM transfer with blended toward steps as fallback; caching and the reduced support-
 364 function lower bound carry over unchanged. Validation: agreement with the exact box-constrained
 365 QP to 10^{-9} – 10^{-14} and with LIBSVM’s NuSVC to $\sim 10^{-10}$ through the mapping $\delta^* = \|w_{\text{svc}}\|/\varsigma$, ς
 366 the per-class sum of the rescaled dual coefficients. The reduced hull is a polytope, so (1) still applies,
 367 but its vertices are blends and the capped transfer is not the pairwise step analysed in §4; the analysis
 368 of this variant is left open.

369 **Kernels.** The algorithm touches data only through inner products, so kernelisation changes plumbing,
 370 not algorithm: Gram columns become kernel columns $\kappa(V, x_i)$, squared norms become kernel
 371 diagonals, iterates exist purely as convex weights (the exact cache refresh is reconstructed from the
 372 support’s cached kernel columns), and the classifier is the kernel expansion of the witness-pair bi-
 373 sector. A ridge $K + I/C$ gives the kernel L_2 margin by the same diagonal trick. Validation: the
 374 linear kernel reproduces the Euclidean solver to 10^{-12} ; RBF distances match the exact kernel QP
 375 to 10^{-9} and LIBSVM’s hard-margin solution, via $\|w_{\mathcal{H}}\|^2 = (\alpha \circ y)^\top K(\alpha \circ y)$ for LIBSVM’s dual
 376 coefficients α and labels y , to 10^{-5} ; kernel SMO matches the QP to 10^{-8} .

377 C Experimental details

378 **Solver settings and tolerance equivalence.** ETA: block transfers with $k = 16$, guarded, away
379 fallback, pairwise anti-zig-zag, stop at certified relative gap 10^{-3} , iteration cap 2×10^6 , 600 s cap.
380 The batteries were run with the unconditional V -then- W order; the schedule of Lemma 7 (larger gap
381 first, skip after a drop) differs from it only on drop iterations, and Table 3 shows the two agree on
382 the synthetic instances of this appendix; screening (every 25 iterations) only in the LIN cell, where
383 it never activated at 10^{-3} , as Theorem 2 predicts. Our SMO: maximal-violating-pair selection, KKT
384 gap 10^{-3} , same caps; hard margin as $C = 10^{12}$; kernel SMO on $K+I/C$. LIBLINEAR (scikit-learn
385 1.8 LinearSVC): squared hinge with parameter $C/2$, so that its objective equals $\frac{1}{2}\|w\|^2 + \frac{C}{2} \sum \xi_i^2$;
386 tolerance 10^{-6} , chosen so that its final primal matches ETA’s certified value on every cell where ETA
387 converges (a9a 687.07 vs 687.07; gisette 0.14900 vs 0.14893, inside ETA’s certified 8×10^{-4} ; ijcn1
388 550.6 vs 550.5 at ETA gap 0.01); this tolerance is also why LIBLINEAR needs 18,000 iterations and
389 over an hour on gisette at $C=1$, and 35,000 iterations and 1.5 h at $C=10$, where it nevertheless stops
390 5% above the certified optimum (0.1566 vs 0.1490). The gisette LIBLINEAR times in Table 2 are
391 therefore the cost of matching ETA’s primal, not LIBLINEAR’s cost at its default tolerance (10^{-4}),
392 which we did not measure; since its dual coordinate descent is monotone, the primal at the default
393 tolerance can only be at or above the 10^{-6} value. The paper claims no wall-clock advantage over
394 LIBLINEAR on gisette. Intercept scaling 100; iteration cap 2×10^5 . LIBSVM (scikit-learn SVC):
395 tolerance 10^{-3} , 1000 MB cache, iteration cap 2×10^6 (reported as *maxiter* when hit); hard margins
396 as $C = 10^6$; RBF with $\gamma = 1/d$. Oracle counts: ETA counts Gram or kernel columns evaluated
397 against a full class ($O(nd)$ or $O(n_k d)$ each); SMO counts kernel rows, the same unit.

398 **Data and seeds.** a9a, w8a, ijcn1 and gisette use the LIBSVM training and test files; covtype
399 (binary, scaled) is subsampled to 100,000 points and split 75/25 by seed. Features are standardised
400 on the training set. Kernel cells use a class-balanced subsample of at most 5,000 points per class
401 (w8a’s minority class has 1,479 points, hence $n_k = 6,479$; gisette uses all 6,000). Seeds vary the
402 covtype split, the kernel subsamples and the timing environment; on datasets with fixed splits the
403 deterministic solvers return identical solutions across seeds, which is why several accuracy CIs are
404 zero. Confidence intervals are t -based 95% intervals over seeds.

ETA on real data (L_2 soft margin, $C = 0.1$): certified gap to the 10^{-3} stopping rule

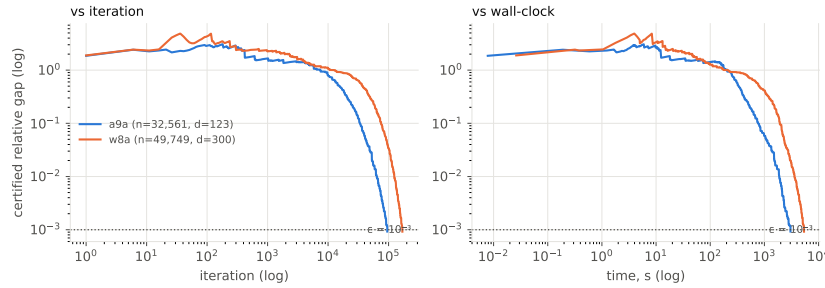


Figure 1: Certified relative gap of ETA on a9a and w8a (L_2 , $C = 0.1$, uncapped), versus iteration and wall-clock: linear-rate convergence in the dense-support regime, at $O(n^2)$ -type cost.

405 **Hardware and concurrency.** One dedicated 32-vCPU Intel Xeon 2.6 GHz Lightning Studio,
406 122 GB RAM, Linux, Python 3.12, NumPy 2.4 with OpenBLAS 0.3.31, OMP_NUM_THREADS=1; the
407 L_2 battery ran 14 cells and the regime battery 12 cells concurrently (the gisette, covtype and w8a
408 $C=10$ cells of Table 2 were completed in later passes with 11–14 workers and no other load). The
409 uncapped a9a/w8a runs in Figure 1 were made earlier on the same machine under a different load
410 (LIBLINEAR took 12 and 48 s there against 4 and 23 s in the capped battery), so ratios across the
411 two runs carry a factor of roughly two of uncertainty; ratios within a battery do not.

412 **Reference solver in the KL2 cells.** The hinge-loss SVC(rbf, $C=1$), a different objective, took
413 8.6 ± 2.9 (a9a), 3.4 ± 0.3 (covtype), 83.5 ± 13.1 (gisette), 1.1 ± 0.0 (ijcn1) and 4.9 ± 1.2 s (w8a),

with test accuracies 0.789, 0.783, 0.977, 0.950 and 0.975, within one point of the L_2 solutions and on either side of them.

Step ablation, blocks and shrinking (synthetic). Iteration counts at $\varepsilon = 10^{-6}$ (single instances): $d=30, n=200$: toward 158,941, midpoint 148,996, away 106, pairwise 181; $d=100, n=1,000$: toward and midpoint fail within 300,000, away 15,886, pairwise 3,921. At the working tolerance 10^{-3} the pairwise/away gain over the midpoint heuristic is only $\sim 11\%$, which is why the heuristic sufficed in [2]. Block transfers cut iterations $4\text{--}30\times$ relative to single MDM (e.g. $43,486 \rightarrow 1,551$ on a dense-support L_2 instance at $d=1,000$) and are the only configuration that reaches the certified tolerance on MNIST odd-vs-even. Figure 2 shows the screening speed-ups and Figure 3 the consolidated comparison: hard margins 0.24 vs LIBSVM’s 0.39 s at $d=1,000$ and 8.6 vs 20.5 s at $d=10,000$; 0.21 vs 0.42 s at $\varepsilon=10^{-5}$ (where the paper-style configuration fails to converge); RBF hard margin 0.76 vs 1.28 s; ν -SVM 18.5 vs 21.9 s; synthetic L_2 ($d=1,000, C=1$) 8.9 vs LIBLINEAR’s 13.8 s; MNIST odd-vs-even 82.8 vs LIBLINEAR’s 55.0 s. Distances agree within tolerance on every row.

Table 3: Two-sided schedules on the synthetic instances of this appendix ($n=5,000$ per class, $k=16$): the unconditional V -then- W order used for the batteries versus the schedule of Lemma 7 (larger-gap side first, second side skipped after a drop). Drop steps and skipped second-side steps are counted over the run.

instance	schedule	time (s)	iterations	drops	skips	distance
hard margin, $d=1000, \varepsilon=10^{-5}$	unconditional	0.1	16	0	0	55.339365
hard margin, $d=1000, \varepsilon=10^{-5}$	Lemma 7	0.2	16	0	0	55.339365
L_2 overlap, $d=100, C=1$	unconditional	7.9	11,216	638	0	0.078400
L_2 overlap, $d=100, C=1$	Lemma 7	8.1	10,701	599	316	0.078400
L_2 overlap, $d=1000, C=1$	unconditional	4.3	2,951	172	0	0.578143
L_2 overlap, $d=1000, C=1$	Lemma 7	4.3	3,001	180	121	0.578143

Table 4: Block-size ablation on the same instances (current code): single MDM ($k=1$) versus blocks of k transfers. Iterations fall with k ; wall-clock does not fall in proportion, because each block iteration touches more columns.

instance	k	time (s)	iterations	columns	drops
hard margin, $d=1000, \varepsilon=10^{-5}$	1	0.3	296	104	0
hard margin, $d=1000, \varepsilon=10^{-5}$	4	0.2	36	103	0
hard margin, $d=1000, \varepsilon=10^{-5}$	16	0.2	16	103	0
hard margin, $d=1000, \varepsilon=10^{-5}$	32	0.2	16	109	0
L_2 overlap, $d=100, C=1$	1	10.7	75,021	1,343	0
L_2 overlap, $d=100, C=1$	4	11.2	24,606	1,632	416
L_2 overlap, $d=100, C=1$	16	9.0	10,701	2,404	599
L_2 overlap, $d=100, C=1$	32	10.4	7,606	3,076	305
L_2 overlap, $d=1000, C=1$	1	12.2	43,521	1,251	0
L_2 overlap, $d=1000, C=1$	4	6.7	10,071	1,330	177
L_2 overlap, $d=1000, C=1$	16	4.3	3,001	1,348	180
L_2 overlap, $d=1000, C=1$	32	4.6	1,741	1,396	170

D Numerical verification of the analysis

Lemma 1 was checked on 4,000 random ensembles including heavily clipped and boundary cases (worst slack at machine precision); Proposition 1 on 500 near-orthogonal ensembles; Lemma 3 to 10^{-9} at arbitrary iterates; Lemma 7 on instrumented two-sided runs (no drop iteration added a support index, and every non-drop iteration met the gain bound with M replaced by the computable upper bound $\text{diam } V + \text{diam } W$); and Theorem 2 on instances whose optimal face was computed by an exact QP: at every activating round all zero-weight non-face points were screened, and the final working set equalled the optimal face exactly. Over 5,000 instrumented live block steps none increased the objective. All checks are test files in the released code.

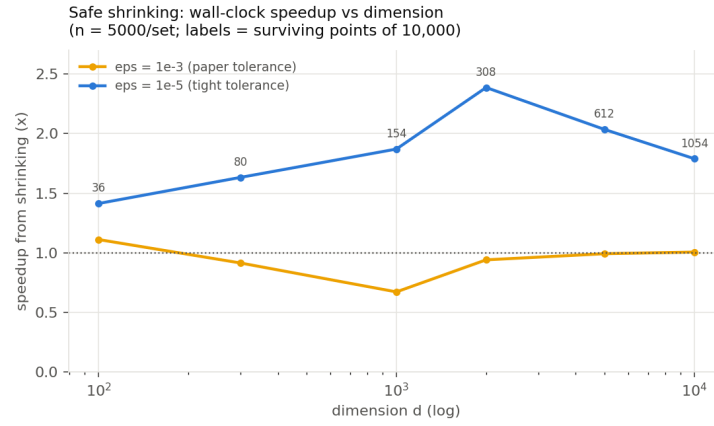


Figure 2: Safe screening: wall-clock speed-up vs dimension at two tolerances on the separated synthetic protocol; labels are surviving points of 10,000.

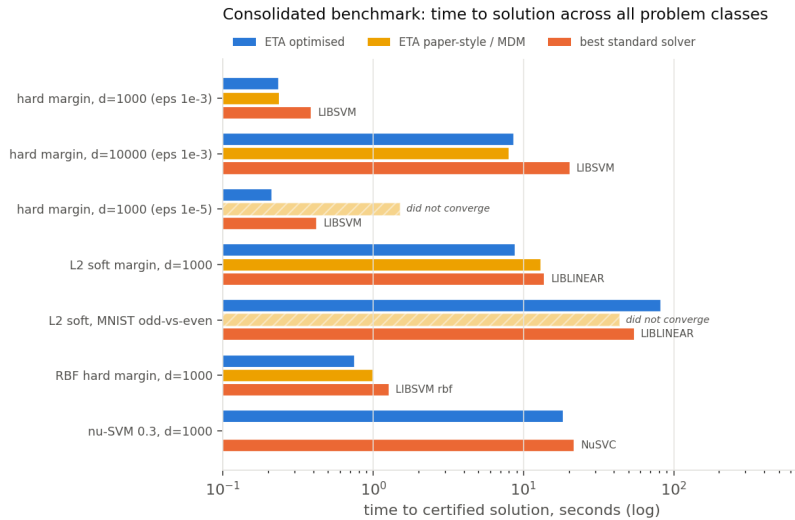


Figure 3: Time to certified solution on seven problem classes: optimised ETA vs the paper-style configuration vs the best standard solver per class.